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Chidaksh Ravuru

Research Interests

Developing reliable agentic orchestration and multimodal foundation models for clinical decision support - specially interested in the alignment, evaluation, and cultural competence of LLMs in high-stakes domains.

Education

2024-Present **University of North Carolina at Chapel Hill**, MS in Computer Science
Advisor: Prof. Shashank Srivastava — GPA: 4.0/4.0

2020-2024 **Indian Institute of Technology, Dharwad**, B.Tech in Computer Science
Advisors: Prof. Prabuchandran K.J. & Prof. Rajshekhar Bhat — GPA: 9.24/10

Selected Publications

2026 **Can LLMs Understand What We Cannot Say? Measuring Multilevel Alignment Through Abortion Stigma Across Cognitive, Interpersonal, and Structural Levels**, *ACM FAccT*
Anika Sharma, Malvika Mampally, **Chidaksh Ravuru**, Kandyce Brennan, Neil Gaikwad

2026 **Unmasking Spurious Correlations to Fix Text Classifiers**, *Ongoing*
Chidaksh Ravuru, Shashank Srivastava

2024 **Agentic Retrieval-Augmented Generation for Time Series Analysis**, *KDD*
Chidaksh Ravuru, Sagar Srinivas Sakhinana, Venkataramana Runkana

2024 **RESTORE: Graph Embedding Assessment Through Reconstruction**, *Preprint*
Hong Yung Yip, **Chidaksh Ravuru**, Neelabha Banerjee, Shashwat Jha, Amit Sheth, Aman Chadha, Amitava Das

Research Experience

Summer 2025 **Applied Scientist Intern**, *Pavo AI*
Designed novel evaluation metrics to capture customer personas into agentic systems
- Curated PersonaBench, a synthetic dataset of 3,000 conversational trajectories, to train and evaluate AI agents for alignment with specific company personas.
- Developed novel evaluation metrics for strategic alignment and used them to fine-tune a reward model, improving an internal planning agent's persona adherence by 10%.

Summer 2023 **Machine Learning Research Intern**, *Tata Research Data and Development Centre*
Implemented multi-agentic orchestration system for time series analysis.
- Developed an Agentic-RAG model for time series, achieving a 12% improvement in prediction accuracy on complex temporal forecasting tasks.
- Boosted model performance by 18% in anomaly detection and 15% in pattern recognition across 400 task-specific benchmarks using finetuned SLMs.

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Projects

2025 **OncoAgentMTB: AI-Powered Molecular Tumor Board**, *MedGemma Challenge*

- Designed a three-stage multi-agent pipeline using Google's HAI-DEF ecosystem: MedGemma 1.5 4B for structured pathology report parsing, MedGemma 27B for multimodal clinical history synthesis, and TxGemma 27B for evidence-based therapeutic recommendations; preliminary evaluation across 10 TCGA cohorts yields 79% somatic variant classification accuracy and F1 of 0.73 on PubMedQA clinical reasoning tasks.
- Automated TCGA genomic-clinical record ingestion and cross-modal data mapping, reducing structured case preparation from ~3 hours to under 30 minutes per patient.

2025 **Persona-Creative Evaluation Pipeline for healthcare creatives**, *Code* | *Report*

- Developed a behavioral evaluation pipeline to study population-level responses to medical creatives across 216+ patient and HCP personas.
- Implemented element-level attribution using GPT-4o to identify clinical triggers and skepticism drivers in pharmaceutical messaging.

2024 **Explainable Brain Tumor Segmentation with U-Nets**, *Code* | *Report*

- Developed BrainSformer, a transformer architecture for 3D volumetric segmentation, achieving a 95.45% Dice score on tumor core regions within the BraTS 2020 dataset.
- Implemented custom Focal and Dice loss functions to handle class imbalance and integrated Grad-CAM++ for clinical explainability.

2024 **Intelligent Planner: Dynamic LLM Query Routing**, *Code* | *Report*

- Architected a dynamic query router using Llama-3.1-8B to orchestrate specialized LLMs, improving inference efficiency and system performance on GSM8K by 3%.
- Engineered the full-stack deployment using Django for backend inference management and React for the frontend.

Teaching

Summer 2025 **Summer School Teacher**, *UNC School of Medicine*
Led a lecture series for high school students on Statistics and Machine Learning (Linear Regression, SVMs) in neuroscience and medical imagery.

2025-Present **Graduate Teaching Assistant**, *UNC School of Data Science and Society*
TA for DATA 110: Intro to Data Science. Conducted lab recitations and taught basic probability, statistics, and Python coding for undergraduates.

2024 **Undergraduate Teaching Assistant**, *Indian Institute of Technology Dharwad*
TA for Introduction to C, Python, and Software Systems Lab; led workshops and delivered guest lectures.

Achievements

2024 Selected as one of the top 1% of applicants countrywide for Google Research Week, Bangalore, India

2023 Secured 4th place in Inter IIT Techfest 2023 in Open Domain Question Answering

2022 Selected for the Machine Learning Summer School (MLSS) 2022 in Krakow, Poland, as one of the few fully funded participants.

Skills

Programming Python, C, C++, JAX, Java, Javascript, Bash/Shell Script, SQL, React, NodeJS, TypeScript, Django, LaTeX

Libraries PyTorch, HuggingFace Transformers, LangChain, LangGraph, LlamaIndex, TensorFlow, PEFT/LoRA, Pandas, NumPy, Scikit-learn

Software Git, Docker, AWS, Azure, Weights & Biases, PySpark, MySQL, Unix/Linux, Databricks

Service & Outreach

Reviewing ACL, EMNLP, ICLR

2021-2023 Appointed as Space Data Science Club Secretary. Directed a team of 30 members, driving data science initiatives focused on space research. Led hands-on workshops on Python and data visualization, training over 50+ students.